

Data Dictionary

Every file, every column, and how to load them.

LPDG INNOVATION HUB — SELECTION CHALLENGE 2026
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Every file, every column, and how to load them. Written out of the data itself, so it cannot go out of date.

What it does **not** tell you is what any of it means for whether a gateway needs a visit. Working that out is the task.

A note on units

Several of these columns are counters read straight off the gateway firmware, and the system does not record what unit they are in. Where the column name says the unit - `offline_duration_sec`, `online_duration_mins` - you can rely on it. Where it does not, this dictionary says "as the gateway reports it" rather than guessing. If a unit matters to what you are doing, the distribution will tell you more than we can.

Why some of it is in German

The network is run with German-built software and it shows in places. A few columns and several values are in German. `kategorie` holds the values `Normal` and `Schlecht`, which mean normal and bad. It is not a puzzle, it is just what the system writes.

Files

File	Format	Grain	Rows
<code>data/telemetry/month=YYYY-MM/part-0.parquet</code>	Parquet, 8 monthly partitions	gateway x hour	1,433,387
<code>data/telemetry_sample_2025-08.csv</code>	CSV	gateway x hour	181,484
<code>data/gateway_master.csv</code>	CSV	gateway	332
<code>data/field_visits.csv</code>	CSV	work order	642
<code>data/meter_read_success.csv</code>	CSV	gateway x week	7,226
<code>data/engineer_review_2026-02.xlsx</code>	Excel	gateway	120

Telemetry covers **2025-08-01 to 2026-03-31**, partitioned by month: 2025-08, 2025-09, 2025-10, 2025-11, 2025-12, 2026-01, 2026-02, 2026-03.

`telemetry_sample_2025-08.csv` is the same telemetry as the `month=2025-08` partition, exported as plain CSV. It is there so you can look at the data without a Parquet reader. The Parquet partitions are the convenient copy and the one to build on.

Loading it

```
import pandas as pd

# All of it. About 1.4 million rows and roughly 50 columns.
telemetry = pd.read_parquet("data/telemetry")

# One month, or a subset of columns, if memory is tight.
jan = pd.read_parquet("data/telemetry/month=2026-01")
part = pd.read_parquet("data/telemetry", columns=["gateway_id", "ts_utc", "reboot_cnt"])
```

DuckDB reads the same directory directly if you prefer SQL:

```
SELECT * FROM read_parquet('data/telemetry/**/*.parquet') LIMIT 10;
```

For reference, measured: the whole telemetry table read from Parquet takes about 0.6 GB of memory. Reading the one-month CSV without specifying column types takes about 0.1 GB. Loading all of it, twice over, peaks around 1.1 GB. So it fits on a normal laptop comfortably - but it is large enough that repeatedly copying the whole table will make your own life slow.

telemetry

One row per gateway per hour. 57 columns.

Column	Type	Description
gateway_id	text	Gateway identifier.
ts_utc	text	Timestamp of the hour, UTC, ISO-8601.
DateDt	text	Calendar date of the hour in the gateway's local time zone (Europe/Berlin).
hour	integer	Clock hour of the day in the gateway's local time zone (Europe/Berlin), 0-23.
rx_nr_pkts	integer	LoRa packets received during the hour.
rx_crc_bad	integer	Received packets that failed their CRC check.
tx_success	integer	Downlink transmissions that completed.
tx_busy	integer	Downlink transmissions abandoned because the radio was busy.
tx_override	integer	Downlink transmissions that pre-empted another.
number_of_messages	integer	Messages handled during the hour.
avg_idletime	decimal	Mean CPU idle time over the hour, as the gateway reports it.
avg_load1	decimal	Mean one-minute load average over the hour.
load1_bigger1	integer	Count of readings in the hour where the one-minute load average was above 1.
load1_bigger2	integer	Count of readings in the hour where the one-minute load average was above 2.
avg_memfree	decimal	Mean free memory over the hour, as the gateway reports it.
avg_uptime	decimal	Mean uptime over the hour, as the firmware reports it.
avg_activeprocess	decimal	Mean count of running processes.
avg_totalprocess	decimal	Mean count of total processes.
reboot_cnt	integer	Reboots during the hour.
reboot_duration_sec	integer	Total time spent rebooting during the hour, seconds.
r_cnt_power_cycle	integer	Reboots attributed to a power cycle.

Column	Type	Description
r_cnt_reboot	integer	Reboots attributed to a software reboot.
r_cnt_unknown	integer	Reboots with no attributed cause.
r_dur_power_cycle	integer	Reboot seconds attributed to a power cycle.
r_dur_reboot	integer	Reboot seconds attributed to a software reboot.
r_dur_unknown	integer	Reboot seconds with no attributed cause.
avg_reboot_duration	decimal	Mean duration of a reboot during the hour, seconds.
reboot_importance	decimal	A score our monitoring system works out for how bad the restarts look.
disconnection_cnt	integer	Backhaul disconnection events during the hour.
offline_duration_sec	integer	Backhaul offline time counter, seconds.
avg_offline_duration	decimal	Mean duration of a disconnection during the hour, seconds.
online_duration_mins	decimal	Minutes online during the hour as the gateway reports them.
no_conn_importance	decimal	A score our monitoring system works out for how bad the dropouts look.
network_2g	integer	Count of readings in the hour where the gateway was attached to 2G.
network_3g	integer	Count of readings in the hour where the gateway was attached to 3G.
network_4g	integer	Count of readings in the hour where the gateway was attached to 4G.
network_unknown	integer	Count of readings in the hour with no attachment reported.
operator_unknown	integer	Count of readings in the hour attached to the unknown network.
operator_3AT	integer	Count of readings in the hour attached to the 3AT network.
operator_A1	integer	Count of readings in the hour attached to the A1 network.
operator_Eplus	integer	Count of readings in the hour attached to the Eplus network.
operator_O2DE	integer	Count of readings in the hour attached to the O2DE network.

Column	Type	Description
operator_OrangeLU	integer	Count of readings in the hour attached to the OrangeLU network.
operator_Salt	integer	Count of readings in the hour attached to the Salt network.
operator_Swisscom	integer	Count of readings in the hour attached to the Swisscom network.
operator_TmobileA	integer	Count of readings in the hour attached to the TmobileA network.
operator_TelekomDE	integer	Count of readings in the hour attached to the TelekomDE network.
operator_VodafoneDE	integer	Count of readings in the hour attached to the VodafoneDE network.
rss_i_good	integer	Count of readings in the hour with RSSI in the good band.
rss_i_normal	integer	Count of readings in the hour with RSSI in the normal band.
rss_i_bad	integer	Count of readings in the hour with RSSI in the bad band.
rscp_rsrp_good	integer	Count of readings in the hour with RSCP/RSRP in the good band.
rscp_rsrp_normal	integer	Count of readings in the hour with RSCP/RSRP in the normal band.
rscp_rsrp_bad	integer	Count of readings in the hour with RSCP/RSRP in the bad band.
ecio_rsrq_good	integer	Count of readings in the hour with Ecio/RSRQ in the good band.
ecio_rsrq_normal	integer	Count of readings in the hour with Ecio/RSRQ in the normal band.
ecio_rsrq_bad	integer	Count of readings in the hour with Ecio/RSRQ in the bad band.

ts_utc is in UTC. dateDt and hour are the same moment written in the gateway's own local time, Europe/Berlin, which is what the original system recorded.

gateway_master.csv

The asset register. One row per gateway.

Column	Type	Description
gateway_id	text	Gateway identifier.
tenant	text	The customer the gateway serves.
site_type	text	Kind of location the gateway is installed in.
region	text	Federal state.

Column	Type	Description
hw_model	text	Hardware model.
antenna_type	text	Antenna fitted at the site.
fw_version	text	Firmware version currently recorded against the gateway.
fw_updated_on	text	Date the firmware was last changed, blank if never.
installed_on	text	Date the gateway was commissioned.
decommissioned_on	text	Date the gateway was taken out of service, blank if still in service.
n_meters_installed	integer	Meters registered behind this gateway.

`meter_read_success.csv`

One row per gateway per week. This comes from the metering system, not from the network.

Column	Type	Description
week_start	text	Monday of the reporting week.
gateway_id	text	Gateway identifier.
meters_expected	integer	Meters that should have been read that week.
meters_read	integer	Meters actually read that week.

`field_visits.csv`

Historic work orders. One row per visit.

Column	Type	Description
visit_id	text	Work order number.
gateway_id	text	Gateway identifier.
requested_on	text	Date the work order was raised.
visited_on	text	Date the technician attended.
reason_reported	text	Reason recorded when the work order was raised. German.
outcome	text	What the technician recorded on site. German.
parts_replaced	text	Part replaced, blank if none. German.
technician_hours	decimal	Hours booked against the work order.

`engineer_review_2026-02.xlsx`

One engineer's review of a set of gateways, carried out on a single day.

Column	Type	Description
gateway_id	text	Gateway identifier.
standort	text	Region and site type, as the engineer recorded them.
Kategorie	text	The engineer's verdict. Normal or Schlecht.

Column	Type	Description
reviewed_on	text	Date of the review.
reviewer	text	Who carried out the review.
Bemerkung	text	The engineer's free-text note. Often blank. German.

predictions.csv — what you submit

Column	Type	Description
week_start	text	Monday of the week, YYYY-MM-DD.
rank	integer	1 to 15 within the week, no repeats.
gateway_id	text	Either identifier format is accepted.
score	decimal	Your ranking score. Any scale.
reason	text	Why this gateway. Written for the operations manager, 300 characters or fewer.

Exactly 15 rows for each of the 8 weeks from 2026-02-02 to 2026-03-23 — 120 rows in total.

Run `python validate_submission.py predictions.csv` before you hand in. It checks the format only, exactly as our marking script does. It says nothing about your answers.